

Adaptive Funnel-Based Stopping-Time Trading Strategy

Updated Roadmap v2 — executive overview



Project: adaptive funnel-based stopping-time trading strategy



Status: ideation complete — ready for execution



Updated: April 2026



Central claim (precise version): Every completed algorithmic exit has nonnegative realised return after transaction costs. This does not prevent mark-to-market drawdowns, indefinite holding, opportunity cost, or forced-exit losses.



Before GPT review (v1)

- No-loss claim too broad
- Z_t treated too much like a fixed-time significance test
- Buy-and-hold benchmark missing
- Mark-to-market risk underemphasized
- Forced-exit distinction unclear
- Core paper and future-work ideas mixed together



After GPT review (v2)

- ✓ No-loss theorem patched with explicit profit threshold h
- ✓ Z_t reframed as a funnel-width / boundary-crossing parameter
- ✓ Buy-and-hold benchmark added
- ✓ Realised wealth and mark-to-market wealth tracked separately
- ✓ Three-layer structure added
- ✓ Limitations section made explicit
- ✓ Practical execution order clarified

Three-layer structure

1



Layer 1 — Core paper (Phases 0–3):

Fixed funnel → honest estimated funnel → rigorous theorem → simulation



2



Layer 2 — Practical extension (Phases 4–6):

Kalman/GARCH adaptive funnel → walk-forward backtest → benchmark



3



Layer 3 — Future work (Phases 7–10):

Regime HMM → randomised stopping → generalised momentum → HJB theory

Core stopping rule

$$X_t = \log S_t$$

$$U(t) = X_{\tau_b} + \sum_{s=\tau_b+1}^t \hat{\mu}_s + k \sqrt{\sum_{s=\tau_b+1}^t \hat{\sigma}_s^2}$$

$$\tau_s = \inf \{ t > \tau_b : X_t > U(t), M_t \leq 0, X_t - X_{\tau_b} > h \}$$

$$h = \log \left(\frac{1 + c_{\text{buy}}}{1 - c_{\text{sell}}} \right)$$



S_t : asset price



$\hat{\mu}_t$: expected return estimate



$\hat{\sigma}_t$: volatility estimate



M_t : hard risk / drawdown guard



τ_s : stopping time (exit time)



$c_{\text{buy}}, c_{\text{sell}}$: transaction costs



Execution path

Phase 1:
theorem + honest
simulation



Phase 2:
realised vs mark-
to-market +
buy-and-hold



Phase 3:
funnel
visualisation



Phase 4:
Kalman μ



Phase 5:
GARCH σ



Phase 6:
walk-forward
real-data backtest

Layer 1 — Core Paper

Phases 0–3: theorem, honest simulation, evaluation, and geometric funnel

This layer forms the main paper. The goal is a clean mathematical core: a no-realised-loss theorem for completed algorithmic exits, honest simulation, and transparent evaluation against buy-and-hold.

1 Phase 0 — Baseline simulation (done)

- Key question:**

Does the core algorithm work at all on simulated data with a fixed known parameter set?

- Price model:**

$$X_{t+1} = X_t + \mu + \sigma \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, 1)$$

$$S_t = \exp(X_t)$$

- Deviation statistic:**
$$Z_t = \frac{X_t - X_{\tau_b} - \mu(t - \tau_b)}{\sigma \sqrt{t - \tau_b}}$$

- Momentum signal:**
$$M_t = X_t - X_{t-\delta}$$



PyQt5 interactive simulation and matplotlib animation already built.

2 Phase 1 — Rigorous theorem + honest simulation



If the sell rule requires $X_{\tau_s} - X_{\tau_b} > h$, with $h = \log\left(\frac{1+c_{buy}}{1-c_{sell}}\right)$, then every completed algorithmic exit has positive realised return after transaction costs.

Core multiplier after costs:

$$\frac{(1 - c_{sell}) S_{\tau_s}}{(1 + c_{buy}) S_{\tau_b}} = \exp(X_{\tau_s} - X_{\tau_b} - h) > 1$$

Honest simulation pipeline:



Generate hidden prices using true μ, σ



estimate $\hat{\mu}, \hat{\sigma}$ from burn-in



run algorithm using estimates only

The theorem covers algorithmic exits, not forced horizon exits.

3 Phase 2 — Realised vs mark-to-market + benchmark

- Wealth definitions:**

$$W_t^m = W_{\tau_b}^r \cdot \left(\frac{S_t}{S_{\tau_b}}\right) \quad \text{while holding}$$

$$W_t^{bh} = W_0 \cdot \left(\frac{S_t}{S_0}\right)$$

- Key metrics:**

- Sharpe ratio
- Calmar ratio
- Max drawdown
- Opportunity cost vs buy-and-hold
- Average holding time
- Number of completed trades

Wealth curves (example view):



Realised wealth (staircase)



Mark-to-market (while holding)



Buy-and-hold (benchmark)

4 Phase 3 — Geometric funnel

$$C(t) = X_{\tau_b} + \sum \hat{\mu}_s$$

$$V(t) = \sum \hat{\sigma}_s^2$$

$$U(t) = C(t) + k \sqrt{V(t)}$$

$$L(t) = C(t) - k \sqrt{V(t)}$$

$$\tau_s = \inf \{ t > \tau_b : X_t > U(t),$$

$$M_t \leq 0, X_t - X_{\tau_b} > h \}$$



Interpret k as a funnel-width / boundary-crossing parameter, not a fixed-time significance level.



What changed from v1 in Layer 1



Explicit profit threshold h added



Realised vs mark-to-market wealth separated



Buy-and-hold benchmark added



Sequential-testing caveat made explicit

Layers 2 and 3 — Practical Extension and Future Work

Adaptive estimation, real-data backtesting, and advanced extensions

Layer 2 turns the core paper into a practical quant workflow. Layer 3 contains strong future-work ideas that should not block the main paper.

Layer 2 — Practical extension (Phases 4–6)

1 Phase 4 — Adaptive μ via Kalman filter

$$\begin{aligned}\mu_t &= \mu_{t-1} + \eta_t, & \eta_t &\sim \mathcal{N}(0, \tau^2) \\ \Delta X_t &= \mu_t + \sigma \varepsilon_t, & \varepsilon_t &\sim \mathcal{N}(0, 1)\end{aligned}$$

$$C(t) = X_{\tau_b} + \sum_{s=\tau_b+1}^t \hat{\mu}_{s|s}$$

- ✓ drift treated as a latent state
- ✓ τ^2 chosen by Kalman log-likelihood on burn-in

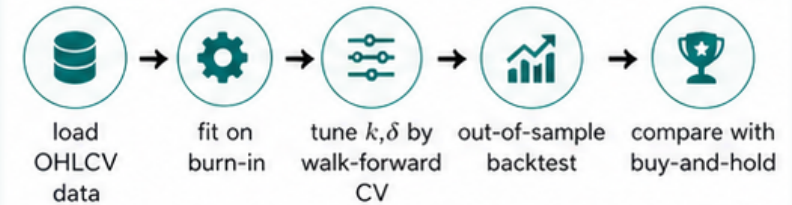
2 Phase 5 — Adaptive σ via GARCH(1,1)

$$\begin{aligned}\Delta X_t &= \mu_t + \sigma_t \varepsilon_t \\ \sigma_t^2 &= \omega + \alpha_G (\Delta X_{t-1} - \mu_{t-1})^2 + \beta_G \sigma_{t-1}^2\end{aligned}$$

$$\begin{aligned}V(t) &= \sum_{s=\tau_b+1}^t \hat{\sigma}_s^2 \\ Z_t &= \frac{X_t - X_{\tau_b} - C(t)}{\sqrt{V(t)}}\end{aligned}$$

- ✓ the funnel breathes with volatility clustering.

Phase 6 — Walk-forward backtest on real data



- ✓ transaction-cost-adjusted h
- ✓ no look-ahead bias
- ✓ report Sharpe, Calmar, max drawdown, opportunity cost

Layer 3 — Future work (Phases 7–10)

1 Phase 7: Regime HMM



online trending vs volatile regime, adaptive k_t

2 Phase 8: Randomised stopping



probabilistic sell rule $p(Z_t)$ above k

3 Phase 9: Generalised momentum



budget/depletion rule, time-varying floor, effective k

4 Phase 10: HJB / free-boundary theory



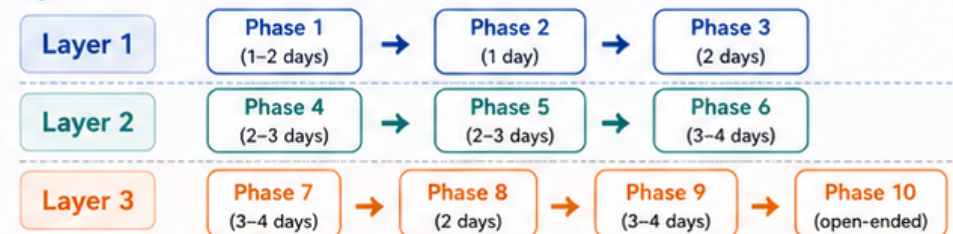
optimal boundary shape and stopping rule

- ✓ Implement in parallel if useful, but do not block the core paper.

✓ Limitations to keep explicit

- mark-to-market drawdowns
- indefinite holding risk
- forced-exit losses
- opportunity cost
- transaction-cost calibration
- model misspecification
- parameter estimation error
- slippage and liquidity
- sequential-testing inflation

Recommended execution order



Adaptive funnel stopping rule

Let log-price be

$$X_t = \log S_t$$

From the buy time, define the predictive centre and cumulative variance:

$$C(t) = X_{\tau_b} + \sum_{s=\tau_b+1}^t \hat{\mu}_s$$

$$V(t) = \sum_{s=\tau_b+1}^t \hat{\sigma}_s^2$$

Upper and lower funnel boundaries:

$$U(t) = C(t) + k\sqrt{V(t)}, \quad L(t) = C(t) - k\sqrt{V(t)}$$

Sell only when the path exits the upper funnel, momentum weakens, and profit after costs is positive:

$$\tau_s = \inf \{t > \tau_b : X_t > U(t), M_t \leq 0, X_t - X_{\tau_b} > h\}$$

$$h = \log\left(\frac{1 + c_{\text{buy}}}{1 - c_{\text{sell}}}\right)$$

Theorem: no realised loss on completed algorithmic exits

If a trade exits only when the profit threshold is satisfied, then the net wealth multiplier is

$$\frac{(1 - c_{\text{sell}})S_{\tau_s}}{(1 + c_{\text{buy}})S_{\tau_b}} = \exp(X_{\tau_s} - X_{\tau_b} - h) > 1$$

Therefore every completed algorithmic exit has positive realised return after transaction costs.

What the safety net does not remove

- Mark-to-market wealth can fall while holding.
- If $\tau_s = \text{infinity}$, capital can be locked indefinitely.
- Forced horizon exits can realise losses.
- The strategy can underperform buy-and-hold.
- Slippage, liquidity, and model misspecification remain real risks.

5 Mathematical Technique Map

Which mathematical ideas each phase is using

Layer 1 - Core paper: mathematical spine

Phase 0 Discrete-time stochastic process; lognormal / GBM discretization; finite-difference momentum; Monte Carlo simulation.

Phase 1 Stopping time; transaction-cost threshold; realised-profit theorem; LLN/CLT for burn-in estimators; Monte Carlo validation.

Phase 2 Wealth process; mark-to-market valuation; buy-and-hold benchmark; drawdown; Sharpe and Calmar ratios; opportunity cost.

Phase 3 Predictive funnel; boundary crossing / first passage; sequential testing caveat; Fokker-Planck connection as theory extension.

Layers 2 and 3 - adaptive models and future theory

Phase 4 State-space model; Kalman filter; latent drift; Bayesian filtering intuition; innovation likelihood for τ^2 .

Phase 5 Conditional heteroskedasticity; GARCH(1,1); MLE; volatility clustering; cumulative conditional variance.

Phase 6 Walk-forward cross-validation; no look-ahead bias; out-of-sample evaluation; risk-adjusted performance metrics.

Phase 7 Hidden Markov Model; forward algorithm; posterior regime probabilities; Markov switching.

Phase 8 Randomised stopping; probabilistic rule $p(Z_t)$; hazard interpretation; functional optimisation with regularisation.

Phase 9 Path-dependent momentum filter; budget/depletion process; adaptive threshold; finite-difference slope.

Phase 10 Optimal stopping; HJB equation; free-boundary problem; variational inequality / impulse control.

How to read this: Layer 1 proves the clean mathematical core; Layer 2 makes the funnel adaptive and testable; Layer 3 contains advanced theory and optional system extensions.

6 Execution Roadmap

From proof to adaptive real-data backtest

1 Theorem + honest simulation

Patch stopping rule with h , separate true parameters from estimates, validate algorithmic exits.

Math terms: stopping time; realised wealth; transaction-cost threshold; LLN/CLT; Monte Carlo.

2 Realised vs mark-to-market

Track staircase realised wealth, dashed mark-to-market wealth, and buy-and-hold benchmark.

Math terms: portfolio process; mark-to-market valuation; drawdown; Sharpe; Calmar; opportunity cost.

3 Geometric funnel

Make $C(t)$, $V(t)$, $U(t)$, $L(t)$ the central mathematical and visual object.

Math terms: predictive interval; boundary crossing; first-passage probability; sequential testing.

4 Kalman drift

Estimate time-varying μ_t as a latent state; update funnel centre online.

Math terms: state-space model; Kalman gain; innovation; filtering; likelihood.

5 GARCH volatility

Estimate conditional variance; make the funnel breathe with volatility clustering.

Math terms: conditional variance; GARCH(1,1); volatility clustering; MLE.

6 Walk-forward backtest

Tune k and δ out-of-sample; report Sharpe, Calmar, drawdown, and opportunity cost.

Math terms: cross-validation; no look-ahead bias; out-of-sample risk metrics.

Recommended start

Run Phases 1, 2, and 3 in parallel because they share infrastructure. Then add Kalman and GARCH. Treat HMM, randomised stopping, generalized momentum, and HJB as future extensions that should not block the main paper.

Three-Layer Structure

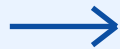
What belongs in the paper vs what becomes future work

Layer 1

Core paper

Phases 0-3

- Baseline simulation
- Rigorous realised-profit theorem
- Realised vs mark-to-market evaluation
- Geometric funnel as central object

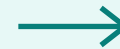


Layer 2

Practical extension

Phases 4-6

- Kalman filter for adaptive μ
- GARCH(1,1) for adaptive σ
- Walk-forward CV and real-data backtest
- Benchmark against buy-and-hold



Layer 3

Future work

Phases 7-10

- Regime HMM
- Randomised stopping rule
- Generalised momentum filter
- HJB / free-boundary theory

?

Best first version: fixed funnel -> estimated funnel -> Kalman/GARCH adaptive funnel -> walk-forward backtest. This gives a coherent research project without drowning the paper in future-work machinery.

8 Limitations and Safeguards

The algorithm changes the form of risk; it does not eliminate risk



Mark-to-market drawdowns

While holding, W_m can decline substantially even if realised wealth remains unchanged.



Indefinite holding risk

If the sell condition is never triggered, $\tau_s = \text{infinity}$ and capital remains locked.



Forced-exit losses

Finite-horizon exits can violate the profit condition and realise losses.



Opportunity cost

Avoiding realised losses can still underperform buy-and-hold in strong trends.



Transaction-cost calibration

The threshold h must reflect commissions, spread, and slippage.



Model misspecification

Real prices have fat tails, jumps, regime shifts, and nonstationary volatility.



Parameter estimation error

Drift is hard to estimate; Kalman filtering helps but cannot remove uncertainty.



Sequential testing inflation

Repeatedly checking Z_t changes fixed-time tail probabilities into boundary-crossing probabilities.

Correct safety-net claim: The algorithm never voluntarily realises a negative return on completed algorithmic exits, provided the explicit profit threshold h is enforced. It still carries mark-to-market, holding-time, and opportunity-cost risk.

9 Implementation Blueprint

Suggested repository structure and deliverables

Repository structure

```
project-trading/  
  core/  
    simulation.py  
    estimation.py  
    funnel.py  
    wealth.py  
  models/  
    kalman_drift.py  
    garch.py  
    regime_hmm.py  
    randomised_rule.py  
    generalised_momentum.py  
  estimation/  
    walk_forward_cv.py  
    momentum_cv.py  
    lambda_optimisation.py  
    hjb_solver.py  
  data/  
    data_loader.py  
    cache/  
  backtest/  
    real_backtest.py  
    benchmark_comparison.py  
    paper_trader.py  
  proofs/  
    proof_check.py  
    estimation_error.py  
  visualisation/  
    trading_interactive.py  
    trading_animation.py  
    funnel_shapes.py  
    funnel_visualiser.py  
  paper/  
    main.tex
```

Core deliverables

- A formal theorem proving positive realised return on completed algorithmic exits.
- An honest simulation where the algorithm uses estimated parameters only.
- Plots of realised wealth, mark-to-market wealth, and buy-and-hold benchmark.
- A geometric funnel visualisation that explains the stopping rule.

Practical extension deliverables

- Kalman filter class with burn-in log-likelihood selection for τ^2 .
- GARCH(1,1) class with online volatility updates.
- Walk-forward cross-validation for k and δ without look-ahead bias.
- Real-data backtest with Sharpe, Calmar, max drawdown, hold time, and opportunity cost.

Before vs After GPT Review

What changed in the roadmap and why it matters

Before GPT review (v1)

- **No-loss claim too broad**
Risked sounding like a risk-free strategy.
- **Z_t treated like fixed-time significance**
Ignored repeated boundary checks over time.
- **Buy-and-hold benchmark missing**
No context for opportunity cost or trend underperformance.
- **Realised vs mark-to-market blurred**
Made the wealth curve look safer than economic reality.
- **Future work mixed with core paper**
Created a kitchen-sink roadmap without a clear spine.

After GPT review (v2)

- **Explicit profit threshold h**
The safety net is now a theorem, not an intuition.
- **Funnel-width interpretation for k**
 Z_t becomes part of a stopping/boundary-crossing system.
- **Benchmark and drawdown added**
Performance is evaluated honestly against alternatives.
- **Algorithmic vs forced exits separated**
The no-loss theorem applies only where it should.
- **Three-layer structure**
Core paper, practical extension, and future work are cleanly separated.

Net result: the project now has a defensible core: adaptive predictive funnel + momentum-confirmed exit + transaction-cost-adjusted no-realised-loss theorem + honest benchmark evaluation.